LI YUNFENG

+86 19857197880 | liyunfeng_lyf27@163.com
Personal website: https://cloud121385.github.io/yunfeng_homepage/

EDUCATION

Zhejiang University Sep. 2023 – Jun. 2027 (Expected)
College of Information Science & Electronic Engineering
Bachelor of Engineering in Microelectronics Science and Engineering
Average score: *86.86 /100.00 (3.92 /4.3)*
Prizes & Awards:
Outstanding Student Scholarship, *Zhejiang University* Sep. 2024
Zhejiang University Five-Star Volunteer, *Zhejiang University* May. 2024

WORK AND INTERNSHIP

Advanced Institute of Information Technology, Peking University
Position: NPU Design Intern Jul 2025 – Jan 2027(Expected)

- Contributed to front-end NPU design across Command Schedule architecture definition, VPU low-bit RTL development, functional verification and debug, and script-based power analysis.
- Benchmarked NVIDIA, Qualcomm, and Huawei Da Vinci scheduling architectures to optimize the existing NPU scheduler, and restructured address-generation logic and RTL to resolve structural and data hazards across SPAD, DDR, and compute-in-memory (CIMC) resources under the existing instruction set.
- Developed and optimized RTL for the multi-format F2M converter, supporting FP32/FP16/BF16-to-MXINT8/MXINT4 conversion, handling quantization boundaries, reusing datapath resources, and tuning PPA based on Design Compiler feedback.
- Built a UVM verification environment with a DPI-C integrated C reference model to cross-check algorithm-level and RTL results, and applied SVA-based formal verification to key control logic and state machines while driving code and functional coverage to closure.
- Automated the Synopsys PrimeTime PX power-analysis flow using Tcl/Python scripts to evaluate switching activity, dynamic power, and peak current under representative workloads; provided analysis data to support clock-gating assessments and power-delivery network design discussions.
- Continuing design refinement and debugging toward the planned November 2026 RTL code freeze and January 2027 tape-out.

ACADEMIC EXPERIENCE

High-Performance GEMM Accelerator Based on Systolic Array Apr. 2026 – Jun. 2026

- Designed a highly parameterized 8x8 Output-Stationary systolic array supporting INT8/INT16 mixed-precision in Verilog, enabling the end-to-end hardware-software co-deployment of the TinyStories-1M large language model on an FPGA.
- Resolved severe timing violations and maximized system Fmax by instantiating DSP IPs and restructuring pipeline registers, while implementing BRAM banking and a Ping-Pong buffer mechanism to overlap data loading with computation for zero-stall streaming.
- Alleviated the Token-decode memory bottleneck by introducing GEMV broadcast and input vector reuse to minimize AXI DMA overhead, and developed Python quantization scripts alongside a MicroBlaze C runtime to handle memory alignment, KV Cache scheduling, and non-linear operators (GELU/LayerNorm).
- Validated the full-stack system on a resource-constrained Nexys4DDR board by establishing a complete PC-UART-FPGA inference pipeline that successfully generated coherent English stories with near-theoretical-limit PE utilization.

RV32I 4-Stage Pipeline Processor and Multi-Level Cache Subsystem Design Apr. 2026 – Jun. 2026

- Developed the full-stack microarchitecture for an RV32I 4-stage pipelined processor and a parameterized multi-level cache subsystem by executing a complete digital IC frontend design workflow from Verilog RTL coding and hardware-software co-verification to FPGA physical synthesis.
- Constructed the core control and data paths by implementing a complex two-stage data forwarding network alongside precise load-use bubble injection and branch prediction flush mechanisms to completely resolve data and control hazards during instruction execution.
- Engineered a multi-level set-associative cache hierarchy featuring a cross-level sub-line bridge and finite state machine models while integrating hardware MMIO performance counters to achieve precise tracking of microarchitectural states including pipeline stalls and cache hit rates.
- Leveraged a custom Python automation framework to efficiently execute regression sweeps across 51 sets of cache parameters for architectural evaluation which quantified a maximum 25.46x architectural speedup under complex algorithmic workloads and successfully met a 100.50 MHz physical implementation timing goal with ultra-low power consumption on the Artix-7 platform.

Monolithic Optoelectronic Chip Based on 12-inch CMOS Silicon Photonics May. 2025 – May. 2026

Position: Principal Investigator

- Led a zero-change monolithic optoelectronic integration project on a domestic 12-inch 55nm (220nm SOI) silicon-photonics platform, perfectly merging the electrical control layer with optical devices (micro-rings/waveguides) on the same top-silicon physical mask layer to successfully complete two tape-outs and the system-level verification of an on-chip micro-spectrometer.
- Innovatively applied side-gate MOSFET structures and built a custom LabVIEW automated testing platform for substrate bias regulation, utilizing Synopsys Sentaurus TCAD and Python fitting algorithms to accurately extract BSIM4 SPICE parameters, achieving high alignment between simulations and fabricated silicon measurements.
- Targeting the demands of large-scale programmable photonic chips, built a highly compatible control circuit IP library from scratch in LTspice, successfully designing a high-precision TIA with 100 dBΩ flat gain, an 8:1 analog MUX, and a 5-bit DAC to effectively suppress charge-injection glitches and optimize settling behavior for nano-ampere signals while meeting all dynamic and closed-loop stability standards.
- Developed a Python-based parameterized layout generation tool using the gdspy library to fully automate non-standard optoelectronic device drawing, reducing layout array iteration time from hours to seconds and ensuring 100% DRC compliance for foundry submission.
- Achieved high-precision separation of 8-channel optical signals within the 1535~1565 nm band with an extreme spectral resolution of 0.04 nm; the corresponding work was accepted for presentation at CIOP 2026, with the conference proceedings currently under IEEE publication review.

Qiushi Wenxin Lab, Zhejiang University

Position: IP Verification Intern Oct. 2025 – Feb. 2026

- Verified and maintained the XFLASHC Flash Controller IP in a UVM environment, analyzing multi-bus routing, dual-bank SWAP address remapping, and security protection mechanisms.
- Performed full-path waveform debugging using VCS and Verdi/Debussy to validate AXI/AHB bus handshakes, register timing sequences, dynamic remapping, and ECC/CRC error handling.
- Enhanced regression test suites for flash operations and error trapping, successfully pinpointing bus hangs and state machine stalls to improve IP verification coverage and robustness.

SKILLS

Programming & Tools: Python (gdspy, PySpice), C/C++, Verilog, MATLAB, LTspice, Synopsys Sentaurus TCAD, Vivado, xsim, LabVIEW

Hardware Design: ASIC/FPGA design flow, RTL design, GDS layout, cache microarchitecture, systolic array architectures, optoelectronic integration

Languages: IELTS 7.5